Henry

1(a)
9
15
5
11
6

(b)

i	a[i] before	total after	a[i] after
0	2	2	2
1	6	8	8
2	5	13	13
3	8	21	21

Output: 8 21

(c) run time error
The program fail to run because num[5] is out of boundary

(d) v is passed by value. In the loop, v is just a copy of value of parameter, so 2*v doesn't change the value of parameter.

Q2(a)

```
FUNCTION countEven(data: ARRAY) RETURNS INTEGER
    DECLARE count, i : INTEGER
    ~~FOR i ← 1 TO LENGTH(data)~~
    count ← 0
    FOR i ← 1 TO LENGTH(data)
        IF data[i] MOD 2 = 0 THEN
            count ← count + 1
        ENDIF
    NEXT i
    ~~RET~~
    RETURN count
ENDFUNCTION
```

Do not write in this margin

(b)
```
DECLARE i, j : INTEGER
~~FOR i ← 1 TO 8~~
DECLARE temp : ARRAY [1:8] OF INTEGER
FOR i ← 1 TO 8
    temp[i] ← INPUT
Next i
~~countEven~~
OUTPUT countEven(temp)
FOR j ← 1 TO 8
    IF temp[j] > countEven(temp) THEN
        OUTPUT temp[j]
    ENDIF
NEXT j
```

14.375
15 18 20 16

(a) [6, 5, 14, 9, 11, 10]
6 9 11

(b) BYREF passes the real address of Data to the Procedure and manages to change the value of Data
~~T~~ Without BYREF, the output will be
3 8 10

(c) ~~PROCEDURE Replace (BYREF Data : ARRAY[1:8] OF INTEGER, OldV~~

(c)
```
PROCEDURE REPLACE(BYREF Data : ARRAY[1:8] OF INTEGER, OldVal : INTEGER, NewVal : INTEGER)
    DECLARE i : INTEGER
    FOR i ← 1 TO 8
        IF Data[i] = OldVal THEN
            Data[i] ← NewVal
        ENDIF
    Next i
END PROCEDURE
```

HENRY

this margi

(d) PROCEDURE (BYREF X : INTEGER, BYREF Y : INTEGER)

~~DECLARE : ox : INTEGER~~

~~ox ← X~~

~~X ← Y~~

~~Y ← ox~~

DECLARE ox : INTEGER

ox ← X

X ← Y

Y ← ox

~~OUTPUT (A & " " & B)~~

ENDPROCEDURE

~~Call Swap~~

DECLARE A, B : INTEGER

A ← 5

B ← 9

Call Swap (A, B)

OUTPUT (A & " " & B)

output: ~~9 5~~ 9 5